

Code-Layout, Readability and Reusability

Date	29 September 2026
Team ID	UNM123APEXLAB
Project Name	ComicCraft - AI Comic Story Creator using Gemini Models
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	4-space indentation followed in all Python files and 2-space in HTML/CSS/JS files.
2	Proper File Structure	Files and folders are logically organized	Yes	Separate folders for routes, services, templates and static files in the FastAPI project.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Names like story_prompt, panel_list and image_url describe their purpose.
4	Function / Method Names	Functions are descriptively named	Yes	Verb-based names such as generate_story_script() and generate_panel_image().
5	Code Comments	Inline and block comments explain logic	Partial	Docstrings and comments added for main functions; a few helper functions still need comments.
6	Modular Design	Code is split into reusable functions/modules	Yes	Gemini calls, authentication and comic rendering are kept in separate modules.
7	No Redundant Code	Duplicate or unused code is removed	Partial	Major duplicates removed; minor unused imports to be cleaned up.
8	Error Handling	Exceptions and errors are handled gracefully	Partial	try/except used for Gemini API calls with user-friendly error messages; more validation planned.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Gemini API Client Wrapper	Python (Google Gemini API)	Story script and image generation modules	High
2	Story Script Generator (prompt builder)	Python (FastAPI)	Story generation and script regeneration	High
3	Panel Image Generation Service	Python (Gemini image model)	Panel generation and single-panel regeneration	High
4	Authentication Module (register, login, email confirmation)	Python (FastAPI)	All protected routes and user pages	Medium
5	Comic Viewer / Layout Component	HTML, CSS, JavaScript	Comic viewer, My Library and PDF export	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	4	Clear folder structure and consistent indentation.
Readability	4	Meaningful names and short, focused functions.
Reusability	4	Gemini, authentication and viewer logic are reusable modules.
Documentation / Comments	3	Main functions documented; helper functions need more comments.
Overall Score	3.75	Average of the four ratings above (out of 5).